

Address Resolution Protocol (ARP)

Outline

Address Mapping

The ARP Protocol

ADDRESS MAPPING

The delivery of a packet to a host or a router requires two levels of addressing: *logical* and *physical*.

We need to be able to map a logical address to its corresponding physical address and vice versa.

These can be done using either *static* or *dynamic* mapping.

ADDRESS MAPPING

Anytime a host or a router has an IP datagram to send to another host or router, it has the logical (IP) address of the receiver.

But the IP datagram must be encapsulated in a frame to be able to pass through the physical network. This means that the **sender needs the physical address of the receiver.**

A mapping corresponds a logical address to a physical address.

- ✓ ARP accepts a logical address from the IP protocol,
- ✓ maps the address to the corresponding physical address and
- ✓ pass it to the data link layer.

Position of ARP in TCP/IP protocol suite

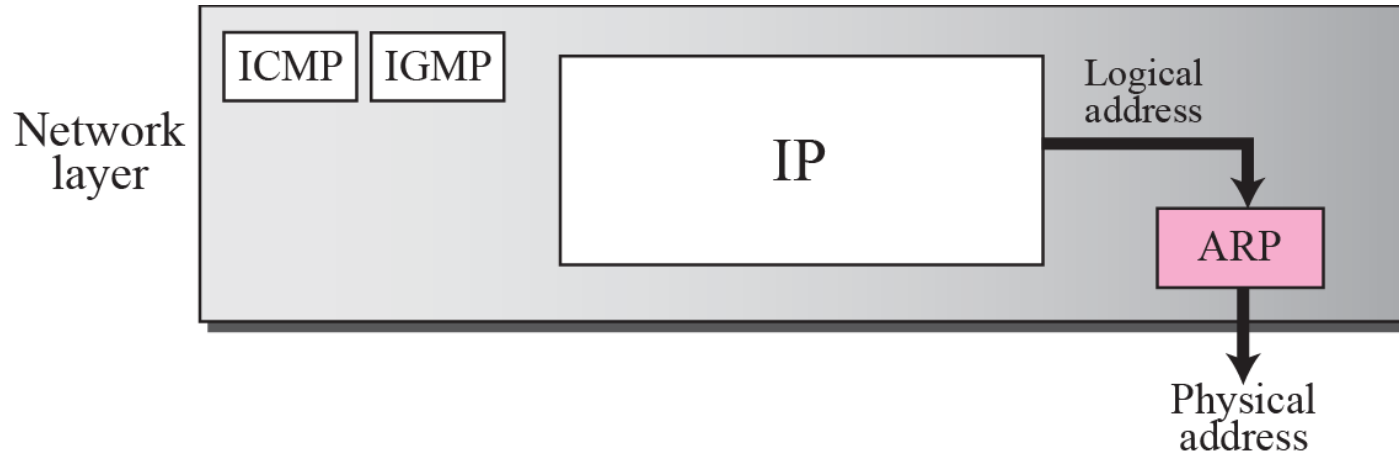
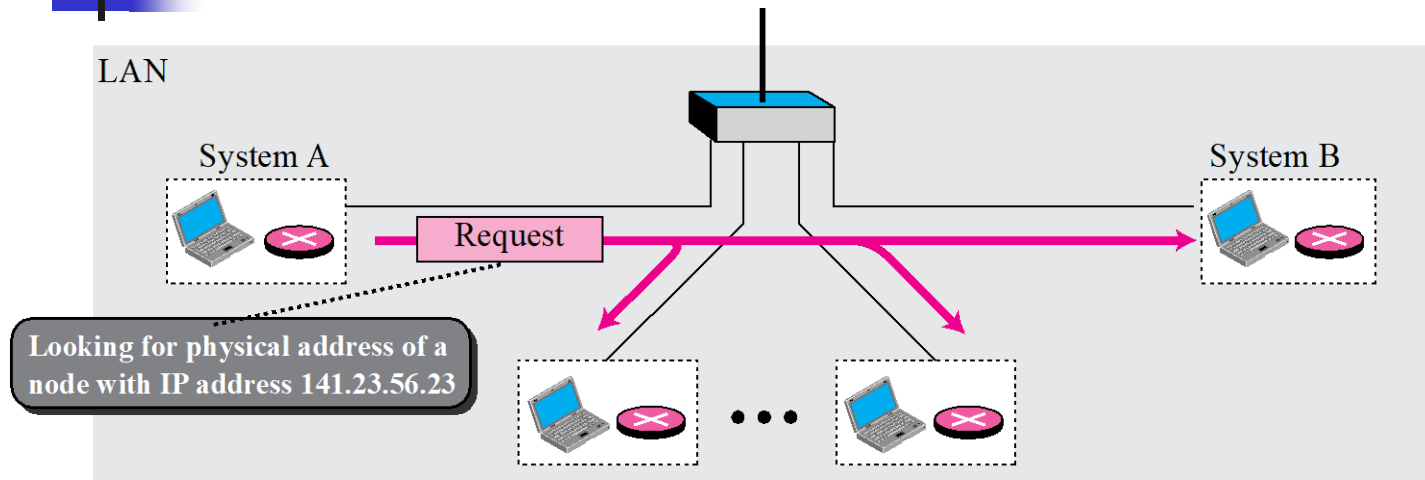
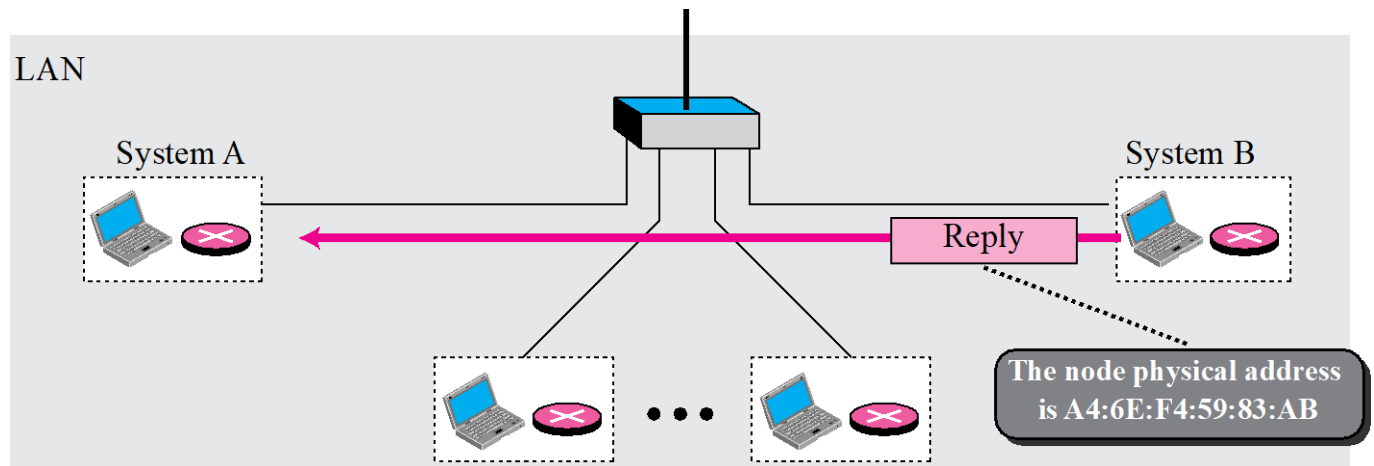


Figure 8.2 *ARP operation*



a. ARP request is broadcast



b. ARP reply is unicast

Figure 3.2 *Ethernet Frame*

Preamble: 56 bits of alternating 1s and 0s.

SFD: Start frame delimiter, flag (10101011)

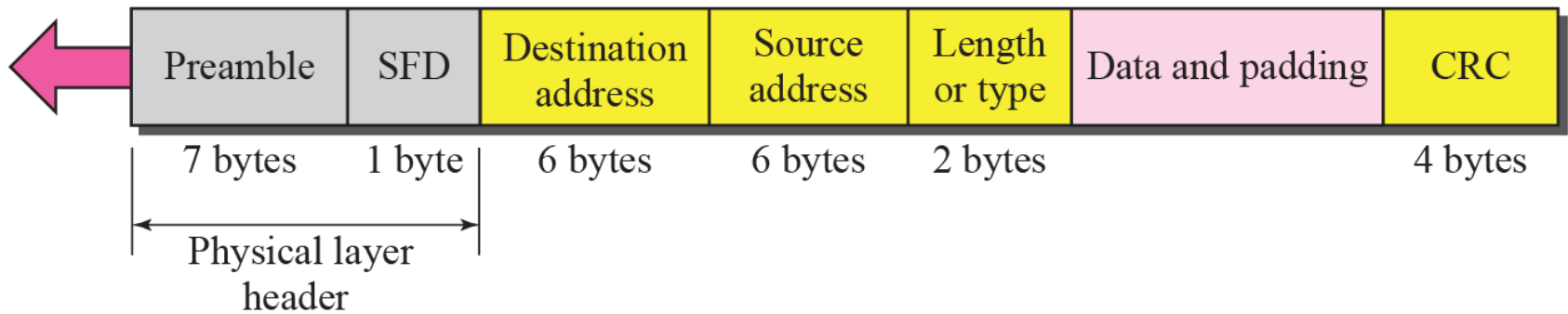
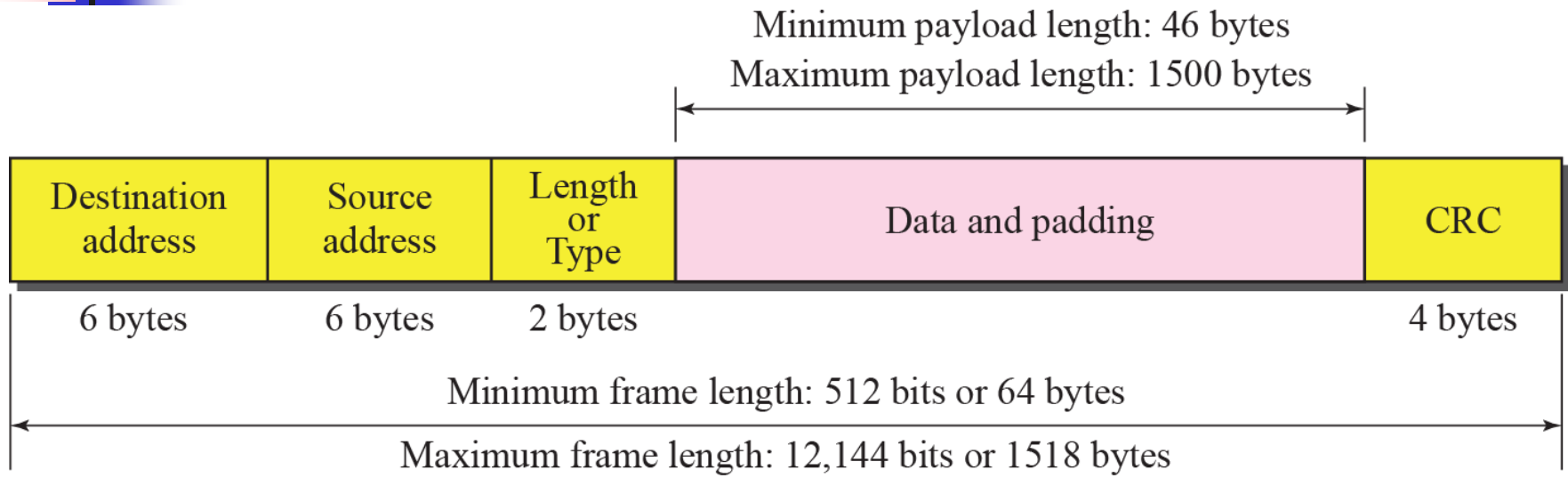


Figure 3.3 *Maximum and minimum lengths*



Minimum length: 64 bytes (512 bits)

Maximum length: 1518 bytes (12,144 bits)

Figure 3.4 *Ethernet address in hexadecimal notation*

d: Hexadecimal digit

d₁d₂ : d₃d₄ : d₅d₆ : d₇d₈ : d₉d₁₀ : d₁₁d₁₂

6 bytes = 12 hexadecimal digits = 48 bits

unicast: 0 multicast: 1





***The least significant bit of the first byte
defines the type of address.
If the bit is 0, the address is unicast;
otherwise, it is multicast.***

***The broadcast destination address is a
special case of the multicast address
in which all bits are 1s.***

Example 3.1

Define the type of the following destination addresses:

- a.** 4A:30:10:21:10:1A
- b.** 47:20:1B:2E:08:EE
- c.** FF:FF:FF:FF:FF:FF



Example 3.1

Define the type of the following destination addresses:

- a. 4A:30:10:21:10:1A**
- b. 47:20:1B:2E:08:EE**
- c. FF:FF:FF:FF:FF:FF**

Solution

To find the type of the address, we need to look at the second hexadecimal digit from the left.

- If it is even, the address is unicast.
- If it is odd, the address is multicast.
- If all digits are F's, the address is broadcast.

Therefore, we have the following:

- a.** This is a unicast address because A in binary is 1010 (even).
- b.** This is a multicast address because 7 in binary is 0111 (odd).
- c.** This is a broadcast address because all digits are F's.

Example 3.2

how the address is sent out on line ?

The transmission is done in following ways

The address is sent

- **left-to-right, byte by byte;**
- **for each byte, it is sent right-to-left, bit by bit,**

Example 3.2

Show how the address 47:20:1B:2E:08:EE is sent out on line.

Solution

- ✓ The address is sent left-to-right, byte by byte;
- ✓ for each byte, it is sent right-to-left, bit by bit,

Ex. 4 7 -> 0100 0111 will be sent as 1110 0010

as shown below:

← 11100010 00000100 11011000 01110100 00010000 01110111

Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

Hardware Type(16)-

define the type of network on which ARP is running. Each LAN is assigned a number based on its type

Protocol type (16) –

defines the protocol. The value of this field for IPv4 is 0800_{16} . It can be used with any higher level protocol.

Hardware length(8) –

length of physical address in bytes.
for example... Ethernet value is 6.

Protocol length (8) –

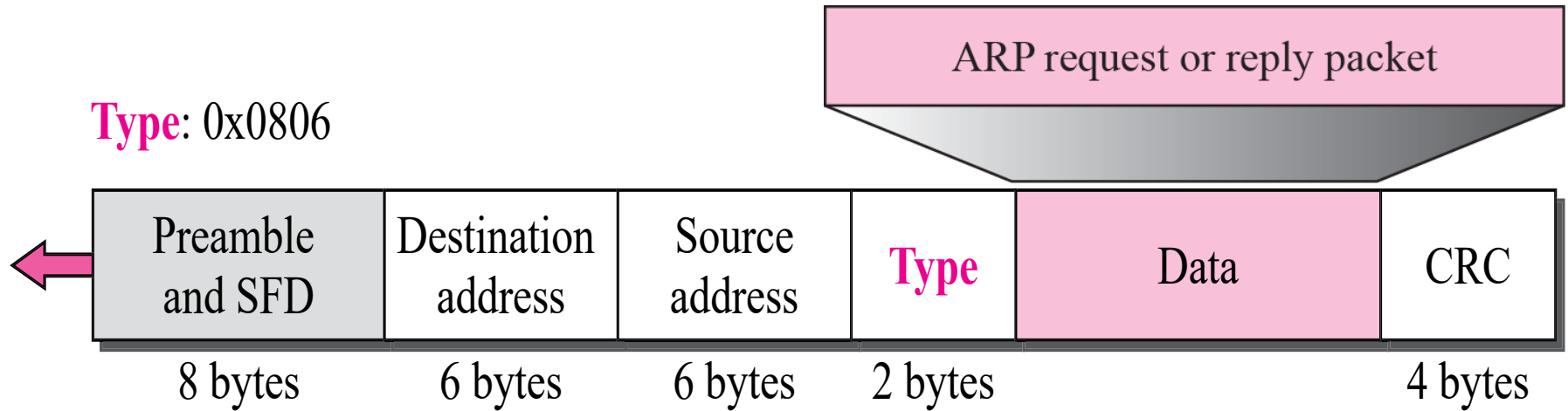
defines the length of logical address in bytes.
Ex for IPv4 protocol the value is 4

Steps Involved

These are seven steps involved in an ARP process:

1. The sender knows the IP address of the target.
2. IP asks ARP to create an ARP request message, filling in the sender physical address, the sender IP address, and the target IP address. The target physical address field is filled with 0s.
3. The message is passed to the data link layer where it is encapsulated in a frame using the physical address of the sender as the source address and the physical broadcast address as the destination address.
4. Every host or router receives the frame. Because the frame contains a broadcast destination address, all stations remove the message and pass it to ARP. All machines except the one targeted drop the packet. The target machine recognizes the IP address.
5. The target machine replies with an ARP reply message that contains its physical address. The message is unicast.
6. The sender receives the reply message. It now knows the physical address of the target machine.
7. The IP datagram, which carries data for the target machine, is now encapsulated in a frame and is unicast to the destination.

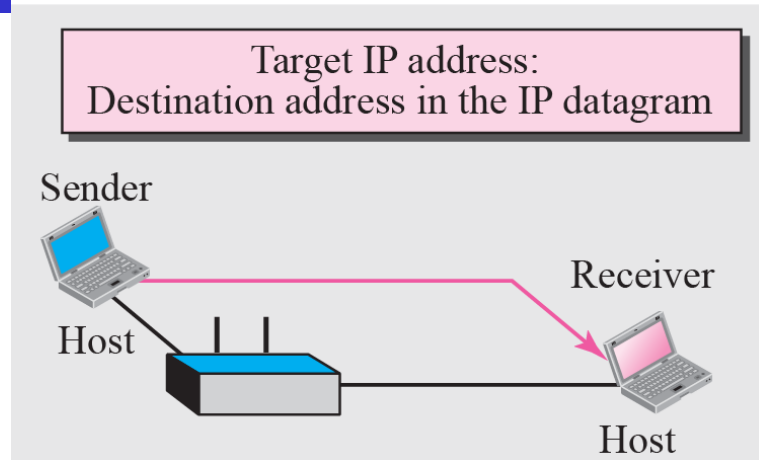
Encapsulation of ARP packet



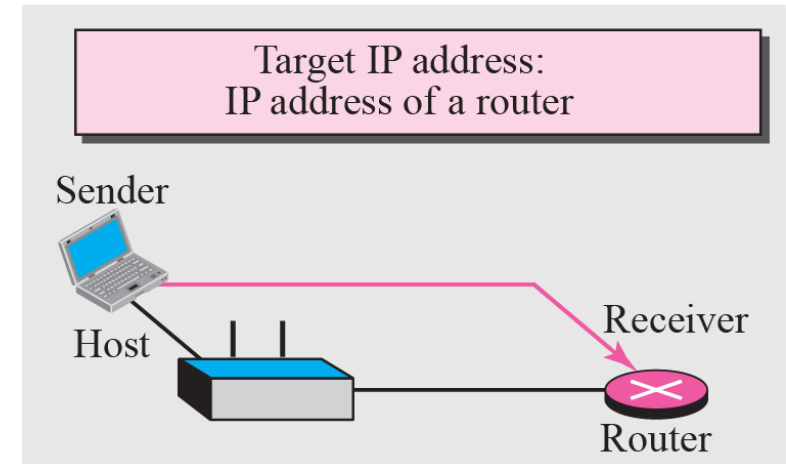
***An ARP request is broadcast;
an ARP reply is unicast.***

Four cases using ARP

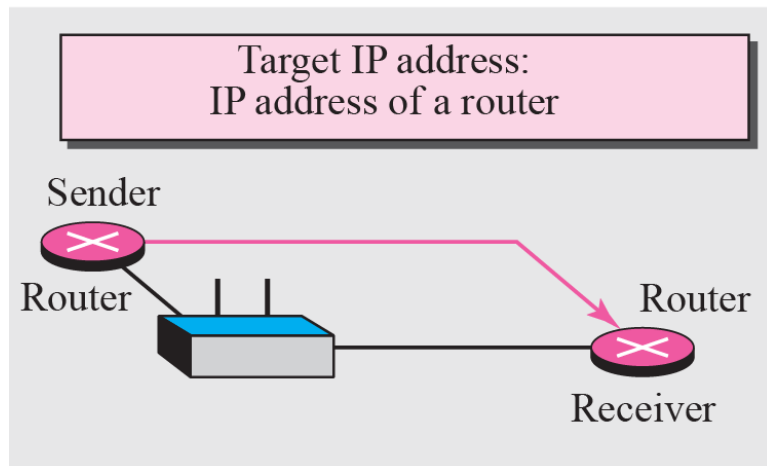
Case 1: A host has a packet to send to a host on the same network.



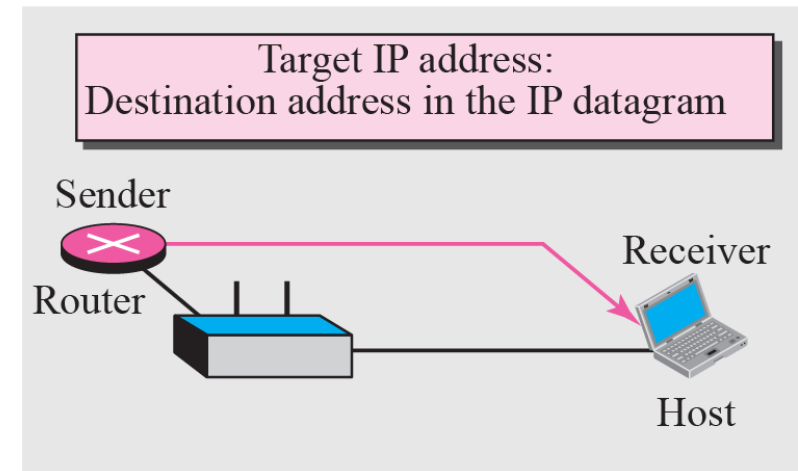
Case 2: A host has a packet to send to a host on another network.



Case 3: A router has a packet to send to a host on another network.



Case 4: A router has a packet to send to a host on the same network.



Example

A host has a packet with
IP address 130.23.43.20 and
Physical address B2:34:55:10:22:10

to send to another host with
IP address 130.23.43.25 and
physical address A4:6E:F4:59:83:AB.

The two hosts are on the same Ethernet network.

Show the ARP request and reply packets encapsulated in Ethernet frames.

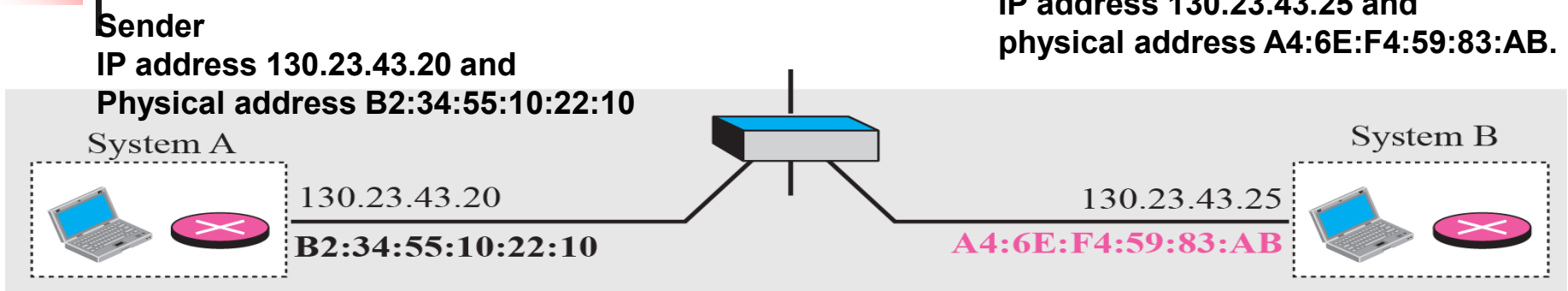
Solution

Figure shows the ARP request and reply packets.
Note that the ARP data field in this case is 28 bytes

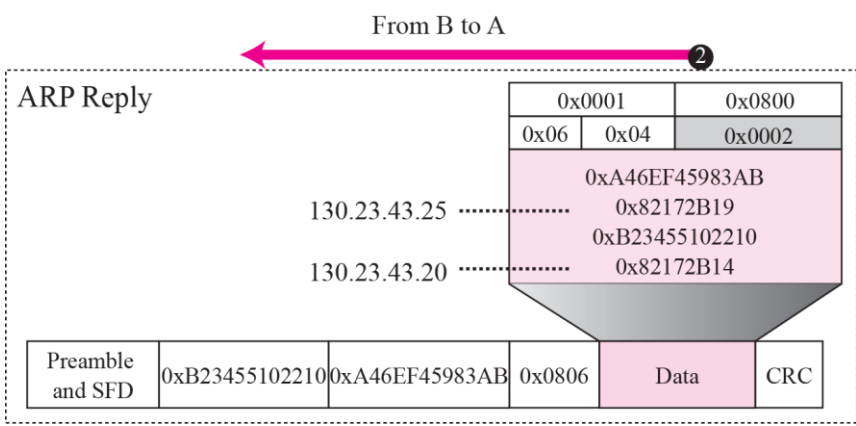
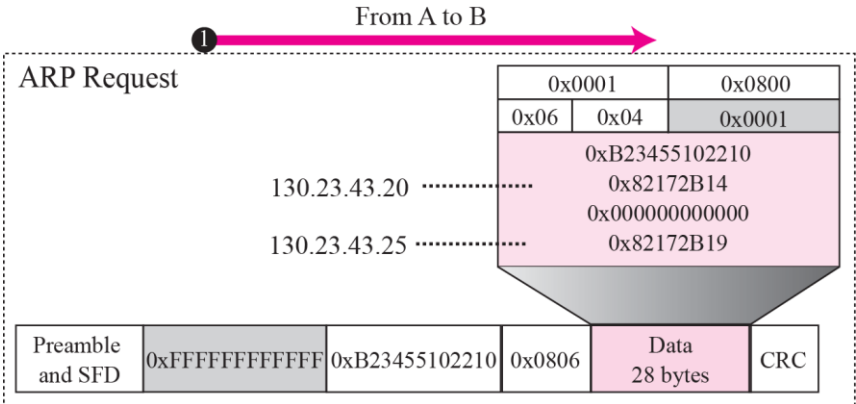
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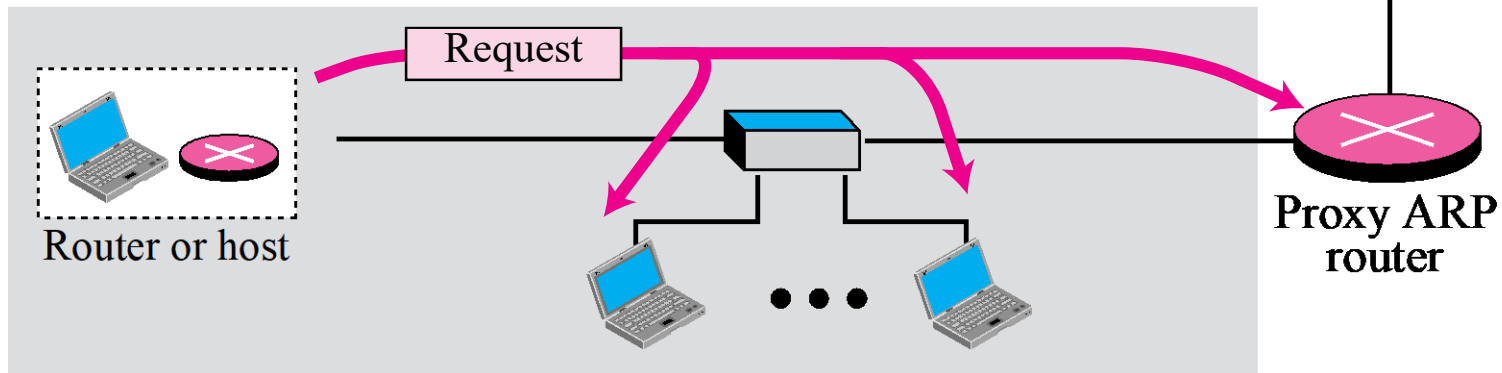
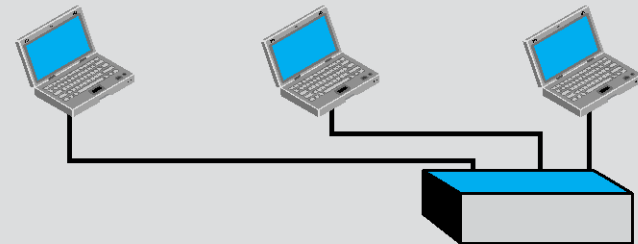
Proxy ARP

It is a technique to create a subnetting effect. A proxy ARP is an ARP that acts on behalf of set of hosts .

The proxy ARP router replies to any ARP request received for destinations 141.23.56.21, 141.23.56.22, and 141.23.56.23.

Added subnetwork

141.23.56.21 141.23.56.22 141.23.56.23



Lecture videos on the topic can be found at

Understanding ARP - Address Resolution Protocol

https://youtu.be/RZPZ_sVzCMU